

PROJECT PROPOSAL

Smart Contract Review System

AI-Powered Contract Clause Rewriting and Comparison

Team Members	Duration
K.V. Mokshith Rao (2024501011)	March - May 2025
G. Hemanth (2024501010)	6 Weeks
K. Anil (2024501059)	Group of 3

1. Project Overview

Contract documents are full of legal clauses that are hard to read and understand for most people. Many clauses are written in a way that is one-sided or unfair, but non-experts cannot always spot this. Our project builds an AI system that reads a contract, finds each clause, rewrites it in a fairer and clearer way, and then shows a side-by-side comparison explaining exactly what changed and why.

The user simply uploads a contract PDF. The system does the rest -- it extracts the text, detects each clause and its type, rewrites it using an AI language model, and compares the original with the rewrite. Everything is shown in a simple web interface.

Note: This system is a decision-support tool only. It does not replace a legal professional, and outputs should be reviewed by a qualified expert before any legal action.

How the System Works:

- Step 1 — User uploads a Contract PDF
- Step 2 — System extracts text from the PDF
- Step 3 — LegalBERT detects and classifies each clause (e.g. Termination, Liability, Confidentiality)
- Step 4 — Each clause is rewritten by an LLM using similar fair clauses retrieved from a corpus (RAG)
- Step 5 — The original and rewritten clause are compared by an LLM, which lists what changed and why
- Step 6 — All results are displayed in the Streamlit web interface

Example:

Stage	Original Clause	After System
Detected Type	"The company may terminate this agreement at any time without notice."	Termination
Rewrite	One-sided -- only the company can terminate, with no notice given.	"Either party may terminate this agreement with 30 days written notice."
Comparison	What changed?	Added notice period Made clause mutual Removed one-sided control

Datasets and Tools:

- CUAD Dataset - 510 real contracts, around 13,000 labeled clause spans across 41 clause types (source: HuggingFace)
- SEC EDGAR - 30 to 50 manually downloaded contracts used as the clause retrieval corpus
- LegalBERT - pre-trained legal language model used for clause detection and classification
- FAISS and SentenceTransformers - used to find similar clauses from the corpus for rewriting
- OpenAI GPT-3.5-turbo or Google Gemini - used for rewriting clauses and generating comparisons
- Streamlit for the web interface, FastAPI for the backend, Google Colab or Kaggle for model training

2. Six-Week Execution Plan

Week	Theme	Member A - ML Lead	Member B - RAG & LLM Lead	Member C - Engineering Lead
1	Data Loading & Setup	Load CUAD, convert to (text, label) format, clean data	Download EDGAR contracts, prepare clause corpus	Set up GitHub repo, PDF/DOCX text converter, FastAPI skeleton
2	Clause Classification	Fine-tune LegalBERT on CUAD for clause type classification	Embed EDGAR clauses using SentenceTransformer, store in FAISS	Connect classification to Streamlit, show clause types on uploaded contract
3	RAG Rewrite Module	Test rewrite outputs, give feedback to Member B	Build rewrite prompt using retrieved clauses and LLM (GPT or Gemini)	Show original clause and rewrite side by side in Streamlit
4	Improvement Explanation	Test and verify explanation outputs are correct	Build explanation module: LLM compares original vs rewrite and lists what changed	Add explanation output to Streamlit, connect full pipeline end to end
5	Testing & Evaluation	Evaluate classification using F1 score, document results	Review 20 rewrites as a team and score on correctness, fairness, clarity	Test full pipeline on 5 contract types, fix bugs, polish UI
6	Report & Demo	Write introduction, data section, classification results	Write RAG module, rewrite module, evaluation, future work	Write system design, conclusion; record demo video; clean GitHub repo

3. Team Roles and Responsibilities

Member	Role	What They Own	Report Section
Mokshith	ML Lead	CUAD data loading and cleaning, converting QA format to (text, label) pairs, LegalBERT fine-tuning for clause classification, model evaluation (F1 score per clause type), comparative benchmark	Introduction, Related Work, Data, Classification Results, Benchmark, Error Analysis
Hemanth	RAG & LLM Lead	EDGAR corpus preparation and chunking, FAISS retrieval setup, rewrite prompt design and testing, LLM-based rewrite generation, improvement explanation module, team-based rewrite quality review	RAG Architecture, Rewrite Module, Improvement Explanation, Evaluation, Future Work
Anil	Engineering Lead	PDF/DOCX to text conversion, FastAPI backend with all endpoints (/extract, /rewrite), Streamlit UI with clause viewer and rewrite panel, SQLite storage, GitHub repo and README	System Architecture, API Docs, Streamlit UI section, Conclusion, README

4. Expected Outcomes and Evaluation

- Clause Classification: F1 score per clause type on CUAD test set (target above 0.70 on major types)
- Rewrite Quality: Team review of 20 rewrites scored on correctness, fairness, and clarity on a 1 to 5 scale
- Improvement Explanation: Verify that AI-generated bullet points correctly reflect what changed between original and rewritten clause
- Full Pipeline Demo: System works end-to-end on 5 contract types - NDA, SaaS, employment, vendor agreement, and lease
- Deployment: Streamlit web app running locally with a clean GitHub repository, setup instructions, and a short demo video

5. Project Scope and Optional Features

Core features to be built within the 6-week timeline:

- Text extraction from uploaded contract PDF
- Clause detection and classification using fine-tuned LegalBERT
- Clause rewriting using RAG -- retrieve similar clauses and generate improved version with LLM
- Clause comparison -- LLM explains what changed between original and rewritten clause
- Streamlit web interface showing all outputs clearly

Optional features -- to be added only if time permits:

- Risk Scoring -- label each clause as HIGH, MEDIUM, or LOW risk based on clause type and keywords
- Confidence triage -- flag clauses as No Review Needed, Review Recommended, or Lawyer Only
- Attention visualization -- highlight which words in the clause influenced the classification

We look forward to your feedback and guidance on this proposal.